

AIRBNB DYNAMIC PRICING ANALYSIS

Minor Project Report

Submitted By: C Nagadeekshika

Internship Program

Submission Date: 30-06-2026

Table of Contents

Type:

1. Introduction
2. Project Objective
3. Dataset Description
4. Data Cleaning
5. Exploratory Data Analysis
6. Dashboard Development
7. Key Findings
8. Conclusion
9. Tools Used

1. Introduction

Airbnb is an online marketplace that connects property owners with travellers seeking short-term accommodation. The platform offers various types of listings such as entire homes, private rooms, and shared rooms.

This project focuses on analysing Airbnb listing data to understand pricing patterns, room type distribution, neighbourhood trends, review behaviour, and availability. The analysis helps identify factors that influence listing prices and provides valuable business insights.

2. Project Objective

The main objectives of this project are:

- To analyse Airbnb listing prices across different room types.
- To compare pricing among neighbourhood groups.
- To study review patterns and listing availability.
- To identify the most expensive neighbourhoods.
- To create an interactive Power BI dashboard for visualization and decision-making.

3. Dataset Description

Dataset Name: Airbnb NYC Dataset

Number of Records: 48,895

Number of Columns: 16

Important Columns:

- room_type
- price
- neighbourhood_group
- neighbourhood
- number_of_reviews
- availability_365
- minimum_nights

4. Data Cleaning

The dataset was cleaned using Python in Jupyter Notebook.

The following steps were performed:

- Checked missing values.
- Identified null records.
- Verified data types.
- Removed unnecessary records.
- Prepared the dataset for visualization in Power BI.

Python libraries used:

- Pandas
- NumPy
- Matplotlib

5. Exploratory Data Analysis

Room Type Distribution:

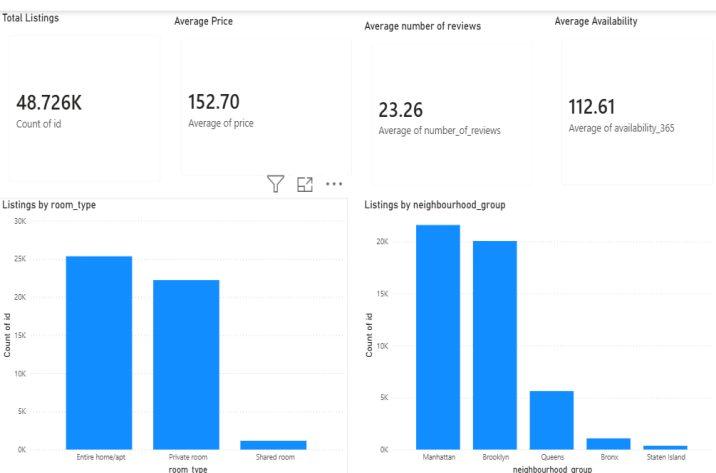
- Entire home/apartment listings were the most common.
- Private rooms formed a significant portion of listings.
- Shared rooms were comparatively fewer.

Average Price by Room Type:

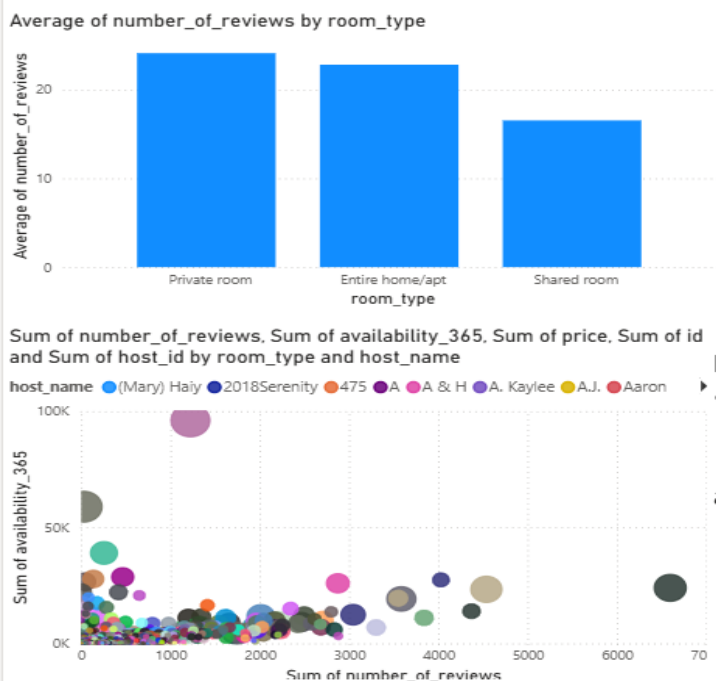
- Entire home/apartment listings had the highest average price.
- Private rooms were more affordable.
- Shared rooms had the lowest average price.

Neighborhood Analysis:

- Manhattan had the highest average listing prices.
- Brooklyn had a large number of listings.
- Several premium neighborhoods showed exceptionally high average prices.



Reviews and Availability Analysis



Key Insights

- Entire home/apartment listings have the highest average prices.
- Manhattan is the most expensive neighborhood group.
- Private rooms provide the most budget-friendly accommodation.
- Price and review count show a weak correlation.
- Availability differs across neighborhood groups.
- Premium neighborhoods command significantly higher prices.

6. Dashboard Development

An interactive Power BI dashboard was developed to visualize pricing, reviews, availability, and neighborhood trends.

The dashboard consists of:

- Overview Dashboard
- Pricing Analysis Dashboard
- Reviews & Availability Dashboard

Interactive slicers were used to filter room types and neighborhood groups.

7. Key Findings

- Entire home/apartment listings have the highest average prices.
- Manhattan is the most expensive neighborhood group.
- Private rooms offer budget-friendly accommodation.
- Price and review count show a weak correlation.
- Availability differs across neighborhood groups.
- Premium neighborhoods command significantly higher prices.

8. Conclusion

The Airbnb Dynamic Pricing Analysis project successfully explored pricing patterns, room types, review behaviour, and neighborhood trends.

The analysis revealed that room type and location play a significant role in determining listing prices. The Power BI dashboard provides an interactive platform for understanding Airbnb market trends and supports data-driven decision-making.

9. Tools Used

- Python
- Jupyter Notebook
- Pandas
- NumPy
- Matplotlib
- Power BI
- GitHub
- Microsoft Word